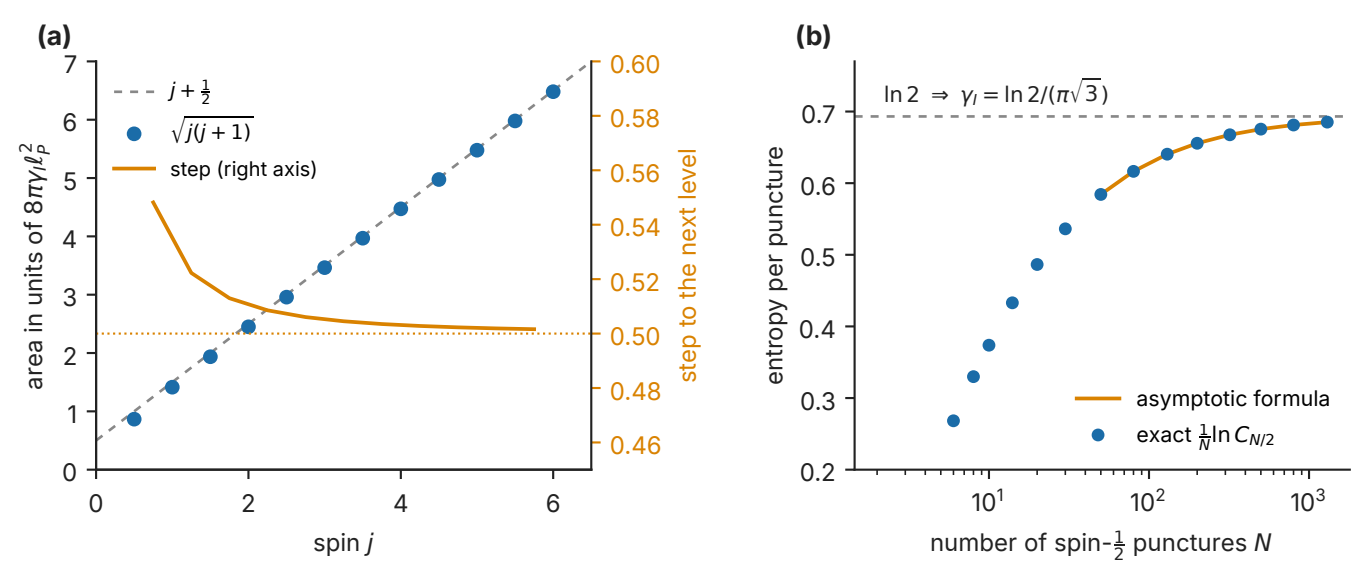
(a)
$j+\frac{1}{2}$
$\sqrt{j(j+1)}$
step (right axis)
area in units of $8\pi\gamma_I\ell_P^2$
step to the next level
spin j
0
1
2
3
4
5
6
7
0.46
0.48
0.50
0.52
0.54
0.56
0.58
0.60
(b)
$\ln 2 \Rightarrow \gamma_I = \ln 2/(\pi\sqrt{3})$
entropy per puncture
asymptotic formula
exact $\frac{1}{N}\ln C_{N/2}$
number of spin-$\frac{1}{2}$ punctures N
0.2
0.3
0.4
0.5
0.6
0.7
10^1
10^2
10^3